(Solution)

FAST-NU
Lahore Campus
Department of Electrical Engineering
Dated: Wednesday, October 29, 2025

EE3003 – Analog & Digital Communication
Fall 2025
Section: BEE-5A
Quiz # 2 on CLO # 02

Name: _____

Roll No.: _____

Total Marks: 15

Time Allowed: 15 minutes

Q1 An angle-modulated signal with carrier frequency $\omega_c = 4\pi \times 10^7$ is described by the equation

$$\varphi_{EM}(t) = 7.5\cos(\omega_c t + 2.5\sin 7000t + 7.5\sin 40{,}000\pi t) \text{ [V]}$$

(a) **Find the signal bandwidth, B in Hz.** [2]

(b) **Find the power of the modulated signal in W.** [5]

(c) **Exposit frequency deviation, Δf in Hz.** [3]

(d) **Find the phase deviation, $\Delta\phi$ in rad.** [3]

(e) **Estimate the bandwidth of $\varphi_{EM}(t)$.** [2]

Write solution below this line. **CLO 02**

(a) $\omega_1 =$ 7000 rad/sec , $\omega_2 =$ 40,000π rad/sec

$f_1 = \frac{\omega_1}{2\pi} = 1114.08\text{Hz}$ $\quad f_2 = \frac{\omega_2}{2\pi} = 20\text{KHz}$

So Signal Band width → $\max(f_1, f_2) = 20\text{KHz}$

(b) $P = \frac{A^2}{2} = \frac{7.5^2}{2} = 28.125\ W$

(c) $\Delta f = ?$

$\omega_i = \frac{d\theta}{dt} = \omega_c + 17500\cos 7000t + 94248\cos 40000\pi t$

~~$\Delta\omega = 7000$~~ $\quad \Delta\omega = 17500 + 300000\pi$

$\Delta f = \frac{17500 + 300000\pi}{2\pi} = 152785\text{ Hz}$

Scanned with CamScanner

(d) phase deviation $\Delta\phi$:-

$$\theta(t) = \omega_c t + (2.5 \sin 7000t + 7.5 \sin 40000\pi t)$$

$$\Delta\phi = 2.5 + 7.5 = 10 \text{ rad}$$

(e) estimate Bandwidth of $\phi_{FM}(t)$ =

$$= 2(\Delta f + B)$$

$$B_{FM} = 2(15278.7 + 20k) = 345579 \text{ Hz}$$



Scanned with CamScanner